

Magnetic-moment-constrained collisions

Equations, numerical design and verification notes

Project: `sato-morrison`

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Abstract

These notes derive and audit a small fixed-field JAX prototype of magnetic-moment-constrained collisions. Implemented paths include the local linear weak operator, a positive nodal nonlinear entropy step, five analytic vacuum fields, tangent toroidal Hamiltonian transport, and an explicit finite-range surrogate. Independent homogeneous controls include Lorentz and Dougherty evolution, physical three-velocity Landau weak moments, and direct screened binary encounters. A checked finite-range Fourier calculation shows how spatial localization creates a density nullspace and how Gaussian range lifts it with quartic long-wave decay. Geometric identities, finite-grid conservation and the selected toroidal observable refinements are verified; other velocity-tail convergence and physical collision-rate calibration remain separate requirements. No calibrated dipole confinement, universal zero-set prescription, full Landau kinetic evolution, or publication priority is claimed.

1 Scope and equation map

The primary source is Ref. [1]. Its Eq. (35) gives a one-species noncanonical collision structure. Equations (111) and (119) are different levels of description: the former evolves a ten-coordinate pair distribution, while the latter closes two five-coordinate species distributions after a pair-factorization assumption. Equation (181) is a localized linear approximation with a prescribed scalar collision coefficient. This prototype must identify which of these models it implements.

The starting model has a prescribed, nonzero vacuum magnetic field, no magnetic perturbation, a fixed electrostatic reference potential, and one species. The nonlinear local-kernel extension is named separately from literal Eq. (181). No absolute collision rate or laboratory confinement result is established by these calculations.

Source equations	Implementation role
SM25 (35), (39)–(59)	Weak form, boundary conditions, invariants and stationary family.
SM25 (73)–(100)	Guiding-center bracket, vacuum assumption and invariant measure.
SM25 (118)–(119)	Pair closure and multispecies kernel; later stage.
SM25 (139)–(150)	Constrained stationary states and initial magnetic-moment marginal.
SM25 (164)–(183)	Linearization, kernel approximations and the first numerical target.
BS25, Sec. V	Structural comparison to a guiding-center Landau bracket.

2 Coordinates, measures and the bracket

We use SI variables before nondimensionalization. Let u be parallel velocity, $\mu = mv_\perp^2/(2B)$ the magnetic moment, and $\eta = \mu B$ the perpendicular kinetic energy. With $\mathbf{b} = \mathbf{B}/B$,

$$\mathbf{B}^* = \mathbf{B} + \frac{mu}{q} \nabla \times \mathbf{b}, \quad B_\parallel^* = \mathbf{b} \cdot \mathbf{B}^*. \quad (1)$$

For the vacuum backgrounds used in Ref. [1], $\mathbf{b} \cdot \nabla \times \mathbf{b} = 0$, hence $B_\parallel^* = B$. This does not imply $\nabla \times \mathbf{b} = 0$: curvature terms in $\mathbf{B}^*$ generally remain.

In coordinates $z = (\mathbf{X}, u, \mu)$ the gyrophase-independent bracket acts on a covector $a = (\mathbf{a}_X, a_u, a_\mu)$ as

$$\mathcal{J}_\mu a = \begin{pmatrix} (\mathbf{b} \times \mathbf{a}_X)/(qB) + \mathbf{B}^* a_u/(mB) \\ -\mathbf{B}^* \cdot \mathbf{a}_X/(mB) \\ 0 \end{pmatrix}. \quad (2)$$

The Hamiltonian and characteristic equations are

$$E = \frac{1}{2}mu^2 + \mu B + q\Phi, \quad (3)$$

$$\dot{\mathbf{X}} = u \frac{\mathbf{B}^*}{B} + \frac{\mathbf{b} \times (\mu \nabla B + q \nabla \Phi)}{qB}, \quad (4)$$

$$\dot{u} = -\frac{\mathbf{B}^* \cdot (\mu \nabla B + q \nabla \Phi)}{mB}, \quad \dot{\mu} = 0. \quad (5)$$

The signs in (2) must be checked against the cross-field drift in (4), not inferred from a memorized matrix convention.

The phase-space measures are

$$d\Gamma = B d^3X du d\mu = d^3X du d\eta. \quad (6)$$

We follow the article's normalization: $n(\mathbf{X}) = \int f du d\eta$. If F is an ordinary gyrotropic velocity density normalized with d^3v , then $d^3v = (2\pi B/m)dud\mu$ and $f = (2\pi/m)F$. Track this constant, particularly in multispecies comparisons.

The coordinate change $\zeta = (\mathbf{X}, u, \eta)$ has derivative

$$T = \frac{\partial \zeta}{\partial z} = \begin{pmatrix} I_3 & 0 & 0 \\ 0 & 1 & 0 \\ \mu(\nabla B)^T & 0 & B \end{pmatrix}, \quad \mathcal{J}_\eta = T \mathcal{J}_\mu T^T. \quad (7)$$

This constructs the matrix in SM25 (93), equivalently (182), without duplicating its entries by hand. It also gives

$$\mathcal{J}_\eta \nabla_\zeta (\eta/B) = 0, \quad \dot{\eta} = \mu \dot{\mathbf{X}} \cdot \nabla B. \quad (8)$$

Thus a discretization that freezes η during spatial transport is not the magnetic-moment-preserving model in a nonuniform field.

With general coordinates and measure $\rho(z)dz$, conservative divergence means $\rho^{-1}\partial_a(\rho F^a)$, not $\partial_a F^a$. Verify the Liouville identity $\partial_a(\rho \mathcal{J}^{ab}) = 0$. The prototype uses $\rho = B$ in the μ chart and $\rho = 1$ in the article's η chart.

Continuous geometric audit

The implemented analytic fields pass automatic-differentiation checks of $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{B} = 0$, antisymmetry, the magnetic-moment Casimir, the Liouville density, and the Jacobi identity in both charts, at independent off-axis points and for both charge signs. Centered differences of $\mathbf{B}$ show second-order convergence to the automatic derivatives for the rational fields. These sampled identities audit the continuous formulas; they do not prove a discrete Jacobi identity for the collision mesh or positivity of B throughout an unsampled domain. Host validation rejects unsupported fields, nonfinite inputs, zero field, excluded dipole/toroidal sources, nonpositive mass, zero charge and negative moment.

For the vacuum toroidal field $B = C/R$, the characteristic equations give $\dot{R} = \dot{u} = \dot{\mu} = 0$, $\dot{\theta} = u/R$ and $\dot{z} = (mu^2 + \mu B)/(qC)$. Thus annular radial boundaries are tangent to Hamiltonian flow; a periodic z and toroidal angle are compatible with this geometry. The library checks this identity and the uniform electrostatic drift independently of collision evolution.

Normalization is part of the closure

Choose fixed scales L_*, v_*, E_*, t_* and a diagonal coordinate scale matrix S . For $\zeta = S\hat{\zeta}$, $E = E_*\hat{E}$ and $t = t_*\hat{t}$,

$$\hat{\mathcal{J}} = t_* E_* S^{-1} \mathcal{J} S^{-T}. \quad (9)$$

State these scales with every dimensional result. A Euclidean phase-space projector applied to a vector containing spatial velocities, accelerations and energy rates is not defined independently of those scales.

For a purely affine chart change $y = Az$ with time and energy conventions fixed, a pair vector transforms as $d_y = Ad_z$. Its quadratic form is unchanged only if

$$\Pi_y = A^{-T} \Pi_z A^{-1}, \quad d_y^T \Pi_y e_y = d_z^T \Pi_z e_z. \quad (10)$$

Recreating an identity matrix and Euclidean projector in the new chart generally specifies a different kernel. For the position-dependent map (7), compute the two vectors in one common chart before subtracting:

$$\Delta_\eta \phi = T(z) \mathcal{J}_\mu(z) \nabla_z \phi - T(z') \mathcal{J}_\mu(z') \nabla_{z'} \phi'. \quad (11)$$

This is the safe route when storing the state on μ nodes but reproducing a kernel defined in the η chart.

3 Nonlinear collision structure and conservation

For a single species define

$$\Delta \phi = \mathcal{J} \nabla \phi - \mathcal{J}' \nabla' \phi', \quad \xi = \Delta E. \quad (12)$$

A specified kernel must obey exchange symmetry, symmetry of its quadratic form, positive semidefiniteness and the energy degeneracy:

$$\Pi = \Pi^T \succeq 0, \quad \Pi \xi = 0. \quad (13)$$

The source strong form is

$$C[f] = \nabla_z \cdot \left[f \mathcal{J} \int f' \Pi (\mathcal{J}' \nabla' \ln f' - \mathcal{J} \nabla \ln f) dz' \right] \quad (14)$$

in a flat invariant-measure chart. Integration by parts, antisymmetry of $\mathcal{J}$ and exchange of the two labels give

$$\int \phi C[f] d\Gamma = -\frac{1}{2} \iint f f' \Delta \phi^T \Pi \Delta \ln f d\Gamma d\Gamma'. \quad (15)$$

All following equalities require the boundary terms to vanish. The paper states conditions for both the total flux and the Hamiltonian part; a finite numerical box does not satisfy them automatically.

Set $\phi = 1$ in (15) to obtain particle conservation. Set $\phi = E$ and use (13) for collision energy conservation. Since $\mathcal{J}\nabla g(\mu) = 0$ for every smooth g ,

$$\frac{d}{dt} \int f g(\mu) d\Gamma = 0. \quad (16)$$

For closed evolution this is equivalent to conserving the full marginal

$$G(\mu) = \int B f(\mathbf{X}, u, \mu) d^3 X du. \quad (17)$$

It is stronger than conservation of $\int \mu f d\Gamma$ alone.

Define $S[f] = - \int f \ln(f/f_{\text{ref}}) d\Gamma$ with a constant reference scale inside the logarithm. Particle conservation removes that scale from the entropy rate. Equation (15) yields

$$\dot{S} = \frac{1}{2} \iint f f' \Delta \ln f^T \Pi \Delta \ln f d\Gamma d\Gamma' \geq 0. \quad (18)$$

These are structural statements from the model, not a proof of a unique final state. Additional null directions of Π can prevent mixing.

A candidate observable p is conserved by the full kinetic model only if the Hamiltonian flow preserves it and the collision form annihilates Δp . An external dipole field does not imply conservation of all Cartesian momenta. For an axisymmetric model, check canonical angular momentum using the actual kernel and boundaries rather than inferring it from the field name.

4 The fixed-field linear model

Write $f = f_0(1 + h)$, where $f_0 > 0$ is stationary. For the source stationary family,

$$f_0 = A \exp[-\beta E_0 + g(\mu)], \quad \mathcal{J}\nabla \ln f_0 = -\beta \mathcal{J}\nabla E_0. \quad (19)$$

With the fields and kernel fixed, the zeroth-order collision term vanishes. The first-order weak form is

$$\int \phi C_L[f_0 h] d\Gamma = -\frac{1}{2} \iint f_0 f'_0 \Delta \phi^T \Pi_0 \Delta h d\Gamma d\Gamma'. \quad (20)$$

The local scalar approximation in SM25 (181) uses

$$\Pi_0 = D P I_x P \delta^3(\mathbf{X} - \mathbf{X}'), \quad P = I - \frac{\xi_0 \xi_0^T}{\xi_0^T \xi_0}, \quad I_x = \text{diag}(1, 1, 1, 0, 0). \quad (21)$$

Here D is a prescribed constant, and the projector is evaluated in a specified normalized common chart. The two velocity integrals remain after the spatial delta function is used. In a μ chart, the local pair measure contains two velocity Jacobians; do not accidentally retain only one factor of B .

The perturbation norm

$$Q[h] = \frac{1}{2} \int f_0 h^2 d\Gamma \quad (22)$$

satisfies

$$\dot{Q} = -\frac{1}{2} \iint f_0 f'_0 \Delta h^T \Pi_0 \Delta h d\Gamma d\Gamma' \leq 0 \quad (23)$$

for collision-only dynamics. A compatible stationary Hamiltonian flow also preserves Q under suitable boundary conditions.

For self-consistent fields, (20) is not a complete linearization unless the omitted terms have been ordered out. Differentiating the exact energy-degeneracy condition gives

$$\delta\Pi\xi_0 + \Pi_0\delta\xi = 0. \quad (24)$$

The approximations in SM25 (164)–(170) must be carried explicitly into any extension. The fixed-field implementation checks its linear operator against a directional derivative of the nonlinear residual; this does not validate omitted self-consistent-field terms.

5 Uniform-field reduction and exact null modes

This section is an independent derivation from the source Eq. (181). Take a uniform field, a constant reference potential and $f_0 = f_0(u, \eta)$. For $\mathbf{b} = \mathbf{e}_z$,

$$\mathcal{J}a = \left(-\frac{a_y}{qB}, \frac{a_x}{qB}, \frac{a_u}{m}, -\frac{a_z}{m}, 0 \right)^T, \quad \mathcal{J}\nabla E_0 = (0, 0, u, 0, 0)^T. \quad (25)$$

Consequently $\xi_0 = (u - u')\mathbf{e}_z$ in the spatial block, and

$$PI_xP = \text{diag}(1, 1, 0, 0, 0). \quad (26)$$

At $u = u'$ use the continuous limit (26); this prescription does not extend automatically to a general nonuniform field.

Only perpendicular spatial derivatives survive. Let

$$n_0 = \int f_0 du d\eta, \quad \delta n = \int \delta f du d\eta. \quad (27)$$

Two applications of the cross-product matrix give $(\mathbf{b} \times)^2 = -I_\perp$. Substitution into the strong form gives

$$\boxed{C_L[\delta f] = \frac{D}{(qB)^2} \nabla_\perp^2 (n_0 \delta f - f_0 \delta n).} \quad (28)$$

This is spatial mixing of velocity-dependent structure, not ordinary homogeneous velocity-space thermalization.

For a Fourier mode with $\mathbf{k}_\perp = \mathbf{k} - (\mathbf{k} \cdot \mathbf{b})\mathbf{b}$, define

$$\lambda = \frac{Dn_0|\mathbf{k}_\perp|^2}{(qB)^2}, \quad \bar{h} = \frac{1}{n_0} \int f_0 h du d\eta. \quad (29)$$

The collision-only equation is

$$\partial_t h = -\lambda(h - \bar{h}), \quad h(t) = \bar{h} + [h(0) - \bar{h}]e^{-\lambda t}. \quad (30)$$

Thus a density-like perturbation is undamped. If $\mathbf{k}_\perp = 0$, every velocity perturbation is collisionally undamped. A homogeneous anisotropic distribution is therefore not an isotropization benchmark for this model.

With positive velocity quadrature weights w_i , set

$$m_i = w_i f_{0i}, \quad s_i = \sqrt{m_i}, \quad n_0 = \sum_i m_i. \quad (31)$$

The matrix for h and its entropy-scaled form are

$$L_h = -\lambda \left(I - \frac{\mathbf{1}\mathbf{m}^T}{n_0} \right), \quad (32)$$

$$M_v^{1/2} L_h M_v^{-1/2} = -\lambda \left(I - \frac{\mathbf{s}\mathbf{s}^T}{n_0} \right). \quad (33)$$

For $\lambda > 0$ the spectrum has one zero eigenvalue and $N_v - 1$ copies of $-\lambda$. This is an exact finite-quadrature statement when the same n_0 is used throughout. It provides a stringent factor, sign and normalization test.

When parallel streaming is added, $\partial_t h = -ik_{\parallel} u h + L_h h$. A density-like collision null mode need not remain in that nullspace under streaming. Distinguish the collision spectrum from the spectrum of the combined generator. With $\mathbf{k}_{\perp} = 0$ and frozen fields the exact streaming solution is $h(t, u, \eta) = h(0, u, \eta) e^{-ik_{\parallel} u t}$; this is not self-consistent Landau damping.

6 A finite-range surrogate and its exact density branch

This section specifies a new calculation for an explicit surrogate, not a Coulomb scattering tensor or a claim of publication priority. Retain the uniform field and constant-potential assumptions of (25), and replace the spatial delta function in (21) by a nonnegative, even, translation invariant kernel $W(\mathbf{X} - \mathbf{X}')$ with $\int W d^3X = 1$. Use the same uniform $PI_x P = I_{\perp}$ prescription, including its continuous equal- u limit. The weak form then gives

$$C_W[\delta f] = \frac{D}{(qB)^2} \nabla_{\perp}^2 [n_0 \delta f - f_0 (W * \delta n)]. \quad (34)$$

To check the sign and convolution placement, the projected pair difference for h is $(\mathbf{b} \times)/(qB)$ applied to $\nabla_{\perp} h(\mathbf{X}, v) - \nabla'_{\perp} h(\mathbf{X}', v')$. The first self term integrates to $n_0 \nabla_{\perp} h$; the cross term is $\nabla_{\perp} W * \delta n$. The second cross-product operation contributes $-I_{\perp}$, giving (34) with the displayed diffusion sign.

Finite-quadrature proposition. Let $m_i = w_i f_{0i} > 0$, $n_0 = \sum_i m_i$, $s_i = \sqrt{m_i}$, $r(\mathbf{k}) = \widehat{W}(\mathbf{k})$, and $\lambda = D n_0 |\mathbf{k}_{\perp}|^2 / (qB)^2$. In the entropy coordinates,

$$-M_v^{1/2} L_{W, \mathbf{k}} M_v^{-1/2} = \lambda \left[I - r(\mathbf{k}) \frac{\mathbf{s}\mathbf{s}^T}{n_0} \right]. \quad (35)$$

Its exact decay rates are λ on the $N_v - 1$ dimensional velocity-neutral subspace and $\lambda[1 - r(\mathbf{k})]$ on the density direction $\mathbf{s}$. Indeed, $\mathbf{s}^T \mathbf{s} = n_0$, so the normalized outer product is an orthogonal rank-one projector. An even nonnegative kernel has real $r \leq 1$; thus all rates are nonnegative. This proof does not require $r \geq 0$. For a discrete periodic spatial quadrature the identical statement holds with r equal to the Fourier multiplier of the sampled, symmetrically normalized kernel, rather than the continuum multiplier.

For a normalized periodized isotropic Gaussian of covariance $\ell^2 I$, $r = e^{-\ell^2 |\mathbf{k}|^2 / 2}$ and hence

$$\lambda_{\text{density}} = \frac{D n_0 |\mathbf{k}_{\perp}|^2}{(qB)^2} \left(1 - e^{-\ell^2 |\mathbf{k}|^2 / 2} \right) = \frac{D n_0 \ell^2}{2(qB)^2} |\mathbf{k}_{\perp}|^2 |\mathbf{k}|^2 + O(|\mathbf{k}_{\perp}|^2 \ell^4 |\mathbf{k}|^4). \quad (36)$$

A transverse long wave therefore has quartic rather than quadratic density decay. Every finite nonzero Gaussian range lifts the nonzero-transverse-wave local density null mode, but the lifted rate vanishes as ℓ^2 in the local limit. Homogeneous and purely parallel modes remain fully collisionally null. Consequently this calculation does not establish a whole-space spectral gap, or a gap for all modes on a periodic domain containing parallel-only modes. On a fixed finite set of transverse Fourier modes, it establishes the stated positive density rates. The limits $t \rightarrow \infty$ and $\ell \rightarrow 0$ need not commute for the density branch.

The independent NumPy check in `examples/06_range.py` assembles the ordered spatial/velocity pair weak sum, including the factor $1/2$. The Fourier basis is differentiated exactly; the Gaussian is evaluated by positive periodic image sums and normalized with one scalar to retain pair symmetry. Three spatial resolutions, three velocity node counts and three kernel ranges are compared with (35). The velocity weights are normalized once at setup to fixed n_0 ; this verifies the finite-quadrature theorem rather than physical velocity-tail convergence. The result is independent of the production JAX pair implementation.

Literature comparison and novelty limits

SM25 supplies the noncanonical weak structure and its local scalar approximation. SM24 supplies the earlier noncanonical construction [2]. BS25, Eqs. (99)–(103), retains magnetic-moment friction and diffusion through its gyrophase-dependent projection Δ_{gc}^μ , and explains the structural reduction obtained by removing that component and replacing the relative velocity by the reduced Hamiltonian pair velocity. That passage does not identify every scalar/local SM25 approximation with the complete conventional Landau tensor. Our uniform surrogate remains unable to isotropize a homogeneous distribution.

KH17 supplies the discrete-gradient conservation and entropy method; those properties do not supply a coercivity constant. Bressan, Kraus, Maj and Morrison [3] analyze sufficient relaxation conditions, including additional finite-dimensional inequalities and tentative infinite-dimensional extensions. Their framework motivates testing the actual nullspace before assigning asymptotic decay to a constrained maximum-entropy state.

On 7 October 2026, primary-source searches of the SM25 title combined with finite range, nullspace and Gaussian terms, and searches for noncanonical finite-range collisions and magnetic-moment quartic relaxation, found no matching statement of (35)–(36). Local full-text checks covered SM25, SM24, BS25, KH17 and BBQ15. This is a scoped independently checked derivation for the specified surrogate; absence from these searches is not proof of publication priority. No general nonuniform finite-range theorem, Coulomb rate, or physical metastability is inferred.

7 Stationary states and initial constraints

The source stationary family (19) follows from energy degeneracy and the magnetic-moment Casimir. For $g(\mu) = -\beta\gamma\mu$,

$$f_* = A \exp \left[-\beta \left(\frac{1}{2} m u^2 + \mu(B + \gamma) + q\Phi \right) \right], \quad B + \gamma > 0. \quad (37)$$

Integrating over u and μ using (6) gives

$$n_*(\mathbf{X}) = A \sqrt{\frac{2\pi}{\beta m}} e^{-\beta q\Phi} \frac{B}{\beta(B + \gamma)}. \quad (38)$$

The corresponding temperature parameters in energy units are

$$T_{\parallel} = \beta^{-1}, \quad T_{\perp} = \frac{B}{\beta(B + \gamma)}. \quad (39)$$

These formulas are useful checks of velocity integration and the definition of pressure. They do not establish that an arbitrary initial state evolves to this particular family member.

For a prescribed fixed-field energy and a known initial marginal $G_0(\mu)$, let

$$Z_{\mu}(\beta) = \int B(\mathbf{X}) e^{-\beta E(\mathbf{X}, u, \mu)} d^3 X du. \quad (40)$$

A candidate constrained maximum-entropy state is

$$f_*(\mathbf{X}, u, \mu) = \frac{G_0(\mu)}{Z_{\mu}(\beta)} e^{-\beta E(\mathbf{X}, u, \mu)}. \quad (41)$$

Solve for β from total energy. Add other invariant multipliers if they actually survive the chosen dynamics. The same initial inverse-temperature parameter need not equal the final multiplier. Extra kernel invariants can make (41) inaccessible.

At fixed G_0 , differentiation of the mean energy of (41) gives

$$\frac{dE_*}{d\beta} = - \int G_0(\mu) \text{Var}_{\beta, \mu}(E) d\mu \leq 0. \quad (42)$$

It is strictly negative if at least one populated moment bin has nonconstant energy. Thus an energy target within the attainable bracket selects a unique β in that bracket. Positive-temperature existence is not automatic for an arbitrary truncated initial distribution; the implementation rejects an unbracketed target. The same calculation holds for positive finite quadrature. With matching energy and full moment marginal, relative entropy gives $S[f_*] - S[f] = \int f \ln(f/f_*) d\Gamma \geq 0$. This identifies a constrained maximum-entropy candidate; it does not establish dynamic accessibility under an operator possessing additional invariants.

For self-consistent electrostatics,

$$-\epsilon_0 \nabla^2 \Phi = \sum_s q_s n_s + \rho_{\text{fixed}}. \quad (43)$$

The self-field energy is $\epsilon_0 \int |\nabla \Phi|^2 d^3 X / 2$, with boundary terms treated consistently. A prescribed external potential contributes $\sum_s \int q_s \Phi_{\text{ext}} f_s d\Gamma_s$ without the self-interaction half-factor. A periodic field solve requires net neutrality and a gauge choice.

8 Comparison models and what each tests

The conventional reduced Landau literature retains magnetic-moment friction and diffusion even when the distribution is gyrophase independent [4, 5, 6, 7]. Brizard and Sugama explicitly discuss a structural reduction with $\Delta_{\text{gc}}^{\mu} = 0$ and a reduced relative velocity in Sec. V of Ref. [7]. This is a bridge for comparison, not a verification that every simplified Sato–Morrison kernel is identical to their Landau tensor. The papers use different conventions; convert all masses, charges and unit factors before comparing implementations.

Pitch-angle scattering

With $v = |\mathbf{v}|$ and $\xi = u/v$, use

$$C_{Lz}[F] = \frac{\nu(v)}{2} \partial_\xi [(1 - \xi^2) \partial_\xi F]. \quad (44)$$

The Legendre identity gives $C_{Lz}[P_\ell] = -\nu\ell(\ell + 1)P_\ell/2$ for constant ν on a speed shell. Hence degrees 0, 1, 2, 3 have rates 0, ν , 3ν , 6ν . Speed-shell population and energy are conserved, but momentum of the evolved species can be transferred to the fixed reservoir. Since $\mu = mv^2(1 - \xi^2)/(2B)$, this is a moment-changing comparison. Derive its coordinate transform; it is not bare diffusion at fixed u in μ .

A nonlinear conserving Fokker–Planck model

The single-species Dougherty form used as a comparison is

$$C_D[F] = \nu \nabla_v \cdot [(\mathbf{v} - \mathbf{U})F + \theta \nabla_v F], \quad \mathbf{U} = \frac{1}{n} \int \mathbf{v} F d^3v, \quad \theta = \frac{1}{3n} \int |\mathbf{v} - \mathbf{U}|^2 F d^3v. \quad (45)$$

Integration by parts shows number and momentum conservation. The trace of the centered covariance is constant. Its matrix satisfies

$$\dot{\Theta} = -2\nu(\Theta - \theta I), \quad \Theta(t) = \theta I + [\Theta(0) - \theta I]e^{-2\nu t}. \quad (46)$$

For an initially Gaussian homogeneous state this specifies the full Gaussian solution. Use it as an exact code test. A fixed bath temperature or drift would define a different model. Neither version is the exact Coulomb operator.

Physical three-velocity Landau reference

A one-species normalized reference is

$$U(\mathbf{w}) = \frac{I - \hat{\mathbf{w}}\hat{\mathbf{w}}^T}{|\mathbf{w}|}, \quad (47)$$

$$C_{La}[F] = \Gamma \nabla_v \cdot \int F F' U(\mathbf{v} - \mathbf{v}') (\nabla_v \ln F - \nabla_{v'} \ln F') d^3v'. \quad (48)$$

Derive the dimensional prefactor for the selected normalization separately. Its positive kernel and velocity-difference null direction support conventional number, momentum, energy and entropy tests. Use an independently assembled small 3V problem before implementing a general transformed operator.

A gyrotropic two-coordinate distribution still has three-dimensional collision geometry. Converge the gyrophase integral or derive its analytic average; a literal 2V Landau kernel is not an equivalent comparator. Maxwell-molecule and BKW-type tests in Ref. [8] can test a different kernel's numerical machinery, but do not establish Coulomb rates.

9 Analytic magnetic fields and boundaries

A vacuum field can be specified locally as the gradient of a harmonic scalar potential. Use the following fields with explicit domains.

Uniform field. $\mathbf{B} = B_c \mathbf{b}_0$, with $|\mathbf{b}_0| = 1$.

Harmonic mirror. Let $R^2 = x^2 + y^2$ and

$$\Psi_B = B_c z + a \left(z^3/3 - zR^2/2 \right), \quad (49)$$

$$\mathbf{B} = \nabla \Psi_B = (-azx, -azy, B_c + a[z^2 - R^2/2]). \quad (50)$$

Its curl and divergence vanish identically. Choose a domain on which B stays strictly positive. The dimensionless variation is controlled by aL^2/B_c . An arbitrary field $B(z)\mathbf{e}_z$ is not an admissible substitute.

Toroidal vacuum field. On $R > R_{\min} > 0$,

$$\mathbf{B} = \frac{C}{R^2}(-y, x, 0). \quad (51)$$

This provides curvature and field-strength variation without a parallel gradient of B . Its noncontractible domain is part of the problem.

Point dipole. Away from the source,

$$\mathbf{B} = \frac{C_d}{r^5}(3xz, 3yz, 2z^2 - x^2 - y^2), \quad \Psi_B = -C_d z/r^3. \quad (52)$$

This is a field test, not a model of a finite winding or material boundary. Exclude $r = 0$ and any near-source region outside the guiding-center ordering.

Symmetry breaking. A controlled addition $\delta\Psi_B = \varepsilon B_c xy/L$ is harmonic. Its gradient can be added to a vacuum background; check the resulting nonzero-field domain and boundary setup.

Do not make a nonperiodic magnetic field periodic by wrapping its array. Begin with collision-only nonuniform tests under natural zero-collision-flux boundary conditions. For full dynamics, require tangent Hamiltonian flux, a derived invariant-compatible reflection, a specified confining potential, or an open system with measured flux budgets. Arbitrary specular reflection need not preserve μ . A density-profile test without these choices is not a closed relaxation experiment.

Complete smooth nullspace of the local toroidal surrogate

The following classification concerns the specified local $PI_X P$ closure, not a conventional Coulomb operator. Assume $B = (C/R)\mathbf{e}_\theta$, $C > 0$, $m > 0$, $q \neq 0$, constant electrostatic potential and $D > 0$. Take a connected radial interval away from $R = 0$, periodic θ, z , and a product velocity domain $I_u \times I_\mu$ with both intervals of positive length. The equilibrium weight is strictly positive in this domain. Consider real C^2 test functions in the weighted form domain, so the pair quadratic integral is finite. Periodicity and natural zero collision flux on nonperiodic faces give the weak identity (20).

Proposition. The smooth collision-null functions are exactly

$$h = \phi(R, \theta, z) + g(\theta, \mu) + a(\theta)E + b(\theta)mRu, \quad E = \frac{1}{2}mu^2 + \mu C/R. \quad (53)$$

Here the functions are arbitrary subject to the stated regularity, periodicity and integrability. Redundant angle-only terms may be moved between ϕ and g . The common collision and ideal stationary functions are exactly

$$h = \phi(R) + g(\mu) + aE + bmRu, \quad a, b \text{ constant}. \quad (54)$$

For a stationary positive f_0 , skew Hamiltonian transport and the collision norm identity imply that these also exhaust the smooth zero modes of the combined linear generator. This classifies stationary null modes; it does not prove a decay rate, coercivity constant, or spectral gap.

Proof of the collision statement. In the common Cartesian η chart, write the spatial action in the local orthonormal basis $(\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z)$:

$$A_X h = \left(\frac{R h_z}{qC}, \frac{h_u}{m}, \frac{-R h_R + u h_u}{qC} \right), \quad V_X = A_X E = \left(0, u, \frac{\mu u^2 + \mu C/R}{qC} \right). \quad (55)$$

The factor m in $\mathbf{B}^*$ cancels the bracket denominator m in the curvature term, so the last numerator contains $u h_u$, not $\mu u h_u$. The energy characteristic has $\dot{u} = \dot{\eta} = 0$; thus its pair vector ξ is spatial-only and $PI_X P$ discards all u, η components. Zero dissipation is consequently equivalent to $A_X h(v) - A_X h(v')$ being parallel to $V_X(v) - V_X(v')$ for all local velocity pairs. Positivity of the pair weight first gives the statement almost everywhere; smoothness extends it throughout the interior.

The radial component gives h_z independent of (u, μ) . The map $(u, \mu) \mapsto (u, (\mu u^2 + \mu C/R)/(qC))$ is invertible onto an open two-dimensional set. On such a set, a map whose every chord is parallel to the corresponding argument chord has the form $F(v) = \alpha v + c$ with scalar α constant on that velocity set. To prove this elementary lemma, fix one point and compare its chords with two noncollinear chords: parallelism of the third chord forces their scalar factors to agree. An open set supplies a noncollinear comparison for each additional point. Applied to the angular and vertical components of (55), this gives

$$\frac{h_u}{m} = \alpha(\mathbf{X})u + c_\theta(\mathbf{X}), \quad \frac{-R h_R + u h_u}{qC} = \alpha(\mathbf{X}) \frac{\mu u^2 + \mu C/R}{qC} + c_z(\mathbf{X}). \quad (56)$$

Integrate the first identity in u to obtain $h = m\alpha u^2/2 + m c_\theta u + d(\mathbf{X}, \mu)$. Comparing the u^2 , u and constant terms in the second yields

$$\alpha_R = 0, \quad (c_\theta)_R = c_\theta/R, \quad d_R = -\alpha\mu C/R^2 - qC c_z/R. \quad (57)$$

Hence $c_\theta = Rb(\theta, z)$ and $h = \alpha(\theta, z)E + b(\theta, z)mRu + g(\theta, z, \mu) + \phi(R, \theta, z)$. The already obtained velocity independence of h_z forces $\alpha_z = b_z = 0$ and makes g_z independent of μ ; its remaining velocity-independent part can be absorbed into ϕ . This proves necessity of (53). The product velocity domain ensures the integration constant is a globally single-valued function of μ ; connected support alone would not exclude branchwise functions on disconnected u -fibers.

For sufficiency, a spatial-only ϕ has velocity-independent spatial action, a $g(\theta, \mu)$ has zero spatial action, an $a(\theta)E$ has pair spatial action $a\xi$, and a $b(\theta)mRu$ has spatial action $b(\theta)R\mathbf{e}_\theta$ independent of velocity. In the last identity, $\mathbf{b} \times (\mu \mathbf{e}_R)/(qB)$ cancels the curvature part from $\partial_u(mRu) = mR$. Their projected pair differences therefore all vanish. This also proves that all spatial populations are conserved by local collisions, including the nonlinear surrogate. Since the local energy-flow map is injective, distinct toroidal velocity states never have $\xi = 0$; the coincident diagonal has vanishing test-function difference.

Proof of the common stationary statement. The ideal derivative is $\mathcal{T}h = (u/R)h_\theta + [(\mu u^2 + \mu C/R)/(qC)]h_z$. Substitute (53) and regard it as a polynomial in u . The cubic coefficient forces $a' = 0$; the constant coefficient $\mu\phi_z/(qR)$ forces $\phi_z = 0$, since the moment interval has nonzero length. The quadratic coefficient then forces $b' = 0$. The remaining linear coefficient is $\phi_\theta + g_\theta = 0$; angle-only redundant terms can be absorbed between ϕ and g , giving (54). Periodic z supplies the closed boundary for the norm argument, but is not needed for this pointwise algebraic intersection. A proposed nonperiodic compensator $\phi = -qCb'(\theta)z$ cancels a quadratic term but leaves $-\mu Cb'/R$, and is not stationary on open moment support.

Thus combined dynamics retain the full radial population, the full moment marginal, energy and toroidal canonical momentum mRu . A general joint (R, μ) marginal is not implied. Arbitrary radial profiles cannot approach a unique state determined only by energy and $G(\mu)$. The same argument classifies positive nonlinear collision equilibria by taking $h = \ln f$ when the entropy form is finite; their common

ideal stationary family is $\ln f = \phi(R) + g(\mu) + aE + bmRu$. Open boundaries with imposed flux, velocity support of dimension one, vanishing pair weights, disconnected velocity fibers, a nonlocal spatial kernel, or an electrostatic field that changes the energy-flow components are outside this proposition.

On the $3^5 = 243$ -node test, 27 spatial indicators, nine angular-moment indicators, three angular-energy functions and three angular-momentum functions have rank 39 after three duplicated angle-only functions are counted once. Their production actions vanish to roundoff and account for all 39 observed near-null eigenvalues of the unmodified matrix. The continuum proof above is independent of that finite-grid rank check; eigenvalues are not altered. A separate continuous-bracket check at $m = 1.7$, $q = -0.8$ and three off-axis points verifies random classified functions and the cylindrical action formula.

Primary-source searches of the authors with toroidal collision nullspace, radial populations and moment constraints, and local review of SM25 and BS25, did not locate this specific local-kernel classification. Local density and symmetry-generated invariants are established ideas; publication priority for this closure-specific classification remains unresolved. No claim about physical Coulomb transport, general magnetic geometry, or a measured lifetime follows from it.

Nonzero-field bounds for the executed domains

For the collision-only Cartesian box $0.8 \leq x \leq 1.2$, $|y| \leq 0.2$, $0.1 \leq z \leq 0.5$, the mirror with $B_c = 1$, $a = 0.15$ has $B \geq B_z \geq 1 + 0.15[0.1^2 - (1.2^2 + 0.2^2)/2] = 0.8905$. For the unit dipole, $B = r^{-3}\sqrt{1 + 3z^2/r^2} \geq (1.2^2 + 0.2^2 + 0.5^2)^{-3/2} > 0.439$. The symmetry-breaking addition with $\varepsilon = 0.03$, $B_c = L = 1$ has norm at most $0.03\sqrt{1.2^2 + 0.2^2} < 0.037$, so the perturbed dipole remains nonzero throughout that box, with $B > 0.402$. These conservative analytic bounds certify the geometry over the domain, beyond sampled minima. They do not certify a guiding-center ordering or a confined Hamiltonian boundary. For the toroidal annulus $1 \leq R \leq 1.5$, $C = 1$, $B \geq 2/3$ exactly.

10 Linear weak discretization and SOLVAX

Represent $h = \sum_i a_i \chi_i$. Define the mass and collision matrices by

$$M_{ij} = \int f_0 \chi_i \chi_j d\Gamma, \quad (58)$$

$$K_{ij} = \frac{1}{2} \iint f_0 f'_0 \Delta \chi_i^T \Pi_0 \Delta \chi_j d\Gamma d\Gamma'. \quad (59)$$

Positive quadrature gives a Gram form $K = B_p^T W_p B_p \succeq 0$. Implement its action without assembling the full pair tensor. A dense assembly on a tiny grid must remain an independent reference.

The executable nodal construction uses tensor differentiation followed by $A_i \phi = T_i \mathcal{J}_{\mu,i} (\nabla_h \phi)_i$ in a common Cartesian η chart. For every unordered pair (i, j) , set $B_{ij} \phi = A_i \phi - A_j \phi$ and construct $\xi_{ij} = B_{ij} e$ from the same discrete energy vector e and derivative. The nonuniform projector uses this discrete ξ , rather than inserting an incompatible continuum energy gradient. This is an explicitly stated discrete energy projector; its consistency must be checked under refinement. The scalar pair weight is $w_X(Bw_v)_i (Bw_v)_j$ locally and $w_i w_j W_{ij}$ for a spatially nonlocal model. In a toroidal chart the spatial weight contains $RdRd\theta dz$ while each local velocity factor contains $Bdud\mu$, even though the one-particle product $BR = C$ simplifies.

The ordered weak form in (59) counts (i, j) and (j, i) equally: exchange reverses both pair differences, leaving their quadratic product unchanged. Therefore its $1/2$ exactly cancels the two ordered contributions, and each unordered pair is inserted once with no $1/2$. Self-node pair differences are identically

zero and need no stored contribution. Distinct nodes with equal energy-flow vectors are a separate issue: the uniform field uses its derived continuous limit, while an exactly zero nonuniform pair vector is rejected. No $|\xi|^2 + \epsilon$ is used. Directional limits may depend on the approach, so successful finite-grid examples do not resolve the general nonuniform zero set.

Periodic derivatives are Fourier derivatives; the real even-grid Nyquist component is unresolved and given zero derivative. Nonperiodic derivatives use local three-point quadratic interpolation, with their transposes in the weak collision assembly. This imposes the natural collision flux rather than wrapping nonperiodic field values. Pair chunks bound transient action workspace; the pair indices and energy directions still require quadratic storage in the resolved velocity count (and spatial count for nonlocal pairs).

If c represents a collision invariant and the derivative/kernel construction respects it, then $Kc = 0$. Include the discrete energy and all resolved magnetic-moment-bin constraints. Finite differences do not automatically obey the product rules needed for an arbitrary sampled μB . Use the same discrete energy gradient in the degeneracy direction and in the weak operator, and prove consistency as the mesh is refined.

The collision-only system and norm balance are

$$M\dot{a} = -Ka, \quad \frac{d}{dt} \frac{1}{2} a^T Ma = -a^T Ka \leq 0. \quad (60)$$

With a compatible stationary Hamiltonian discretization,

$$M\dot{a} = (A - K)a, \quad A^T = -A. \quad (61)$$

Also test $Ac = 0$ for the required linear invariants; skew symmetry alone does not imply those individual identities.

In the nonuniform combined example, exact Fourier half-translations in the periodic toroidal angle and z surround one backward-Euler collision step. The translation uses $\dot{\theta} = u/R$ and $\dot{z} = (mu^2 + \mu C/R)/(qC)$ at each fixed (R, u, μ) . The measure $BR = C$, energy and moment marginal are invariant under this ideal flow, and radial walls are tangent. This symmetric composition remains first order because the collision substep is backward Euler. It is a linear, fixed-equilibrium combined evolution; the nonlinear examples are collision only.

A backward-Euler collision step solves

$$(M + \Delta t K)a^{n+1} = Ma^n. \quad (62)$$

For positive-definite M , this is an appropriate PCG problem. SOLVAX's `pcg_linear_solve` supplies a converged-system differentiation wrapper [9]. Do not pass $M^{-1}K$ to Euclidean PCG without mass scaling or a proof of symmetry. A coupled implicit step with A is generally nonsymmetric and calls for GMRES or another suitable method.

For a linear invariant c , multiply (62) by c^T to check its conservation. Backward Euler also satisfies

$$Q^{n+1} - Q^n = -\Delta t (a^{n+1})^T K a^{n+1} - \frac{1}{2} \|a^{n+1} - a^n\|_M^2 \leq 0. \quad (63)$$

This does not prove positivity of a reconstructed finite-amplitude distribution. An implicit midpoint step can be second order, but is not automatically a good stiff-decay scheme or a nonlinear entropy-preserving integrator.

Local pair application costs $O(N_X N_v^2)$ in a direct quadrature. A spatially nonlocal direct application can cost $O(N_X^2 N_v^2)$. Chunk pair contractions, reuse geometry and quadrature, and keep derivatives sparse. Block-tridiagonal solves may be useful in a line preconditioner; the full velocity coupling is not generally tridiagonal.

11 A nonlinear discrete entropy construction

This finite-dimensional fixed-field construction is implemented. It is not a guarantee for arbitrary interpolation, self-consistent fields, or nonlinear Hamiltonian transport. It follows the weak-structure approach developed for Landau discretizations [10, 11, 12, 13].

Let $f_i > 0$, $w_i > 0$, and $n_i = w_i f_i$. Construct a symmetric positive mobility $K(f)$ from the discrete pair weak form, with

$$K\mathbf{1} = 0, \quad Ke = 0, \quad Kc_\mu = 0, \quad (64)$$

where e is the fixed-field energy vector and c_μ spans the represented moment-marginal constraints. Define

$$S(n) = - \sum_i n_i \ln(n_i/w_i). \quad (65)$$

Then the collision ODE can be written

$$\dot{n} = K(f)\nabla_n S = -K(f) \ln f. \quad (66)$$

The constant term in $\nabla_n S = -\ln f - \mathbf{1}$ vanishes through (64). Consequently $\dot{S} \geq 0$ and $d(e^T n)/dt = 0$. This is an algebraic guarantee only for the actual constructed K .

For each component define a divided difference of $s_i(n) = -n \ln(n/w_i)$:

$$(\bar{\nabla} S)_i = \begin{cases} [s_i(n_i^+) - s_i(n_i^-)]/(n_i^+ - n_i^-), & n_i^+ \neq n_i^-, \\ -\ln(n_i^-/w_i) - 1, & n_i^+ = n_i^-. \end{cases} \quad (67)$$

Evaluate it stably near equal arguments using logarithmic divided differences, not cancellation-prone subtraction. A step

$$n^+ - n^- = \Delta t \bar{K} \bar{\nabla} S, \quad \bar{K} = \bar{K}^T \succeq 0 \quad (68)$$

with the null directions in (64) gives

$$S(n^+) - S(n^-) = \Delta t (\bar{\nabla} S)^T \bar{K} \bar{\nabla} S \geq 0. \quad (69)$$

The conservation laws follow by left multiplication with each invariant vector. Use a positive mean state to construct $\bar{K}$ and a converged nonlinear solve. Parameterizing $n_i^+ = w_i e^{g_i^+}$ enforces positive nodal populations, but does not prove existence of the solve or positivity between interpolation nodes. Check all required quadrature/reconstruction locations.

For the executed nonlinear uniform and linear toroidal examples, continuous spatial Fourier positivity is checked with a sufficient certificate. If $f(x) = \sum_k \hat{f}_k e^{ikx}$, then

$$f(x) \geq \hat{f}_0 - \sum_{k \neq 0} |\hat{f}_k|. \quad (70)$$

The same triangle bound holds for a double Fourier sum in (θ, z) . The example rejects a nonpositive bound, and additionally oversamples the reconstruction. Positive piecewise-linear interpolation in the remaining coordinates then preserves positivity throughout that declared reconstruction. This certificate is conditional on the computed coefficients and is not a positivity theorem for arbitrary Fourier states or higher-order velocity interpolation. The collision derivative still uses its explicitly stated local quadratic differentiation, independently of the positivity-preserving display reconstruction. The Cartesian collision-only boxes use positive nodal values with piecewise-linear reconstruction.

For self-consistent fields the energy is nonlinear in n . Its discrete gradient must enter the collision degeneracy consistently; the fixed vector e above is not enough. Likewise, nonlinear Hamiltonian transport requires its own conservative and positivity-compatible treatment. A linear skew matrix alone is not a proof of a nonlinear H-theorem.

12 Scattering physics and validity tests

The constrained model assumes negligible accumulated change of magnetic moment. That assumption is stronger than a small collision frequency compared with the gyrofrequency. At constant B , an energy change satisfies

$$\Delta E = \frac{m}{2} \Delta(u^2) + B \Delta\mu + q \Delta\Phi. \quad (71)$$

Small ΔE does not separately bound $\Delta\mu$. In particular, the approximate identification in SM25 (85), printed p. 10, requires an extra bound on parallel-energy exchange: at constant B ,

$$B|\Delta\mu| \leq |\Delta E| + \frac{1}{2}m|\Delta(u^2)| + |q\Delta\Phi|. \quad (72)$$

Spatial locality and nearly constant B alone supply no bound on the second term. This audit concerns the inference from small energy change; it does not rule out a stronger regime in which every relevant exchange is ordered small. For independent unbiased increments, zero mean accumulation can coexist with

$$\text{Var} \left(\sum_{j=1}^N \delta\mu_j \right) = N \text{Var}(\delta\mu). \quad (73)$$

For independent increments, both $N|\langle\delta\mu\rangle|/\mu_{\text{ref}} \ll 1$ and $N \text{Var}(\delta\mu)/\mu_{\text{ref}}^2 \ll 1$ are needed for small accumulated change on a chosen nonzero scale. Compensating mean changes can still alter the full moment marginal through diffusion. These are audit points for SM25 (85)–(86), which the source itself calls working assumptions, rather than replacements for its ordering.

A controlled microscopic reference can integrate

$$m_s \dot{\mathbf{v}}_s = q_s \mathbf{v}_s \times \mathbf{B} - \nabla_{\mathbf{r}_s} V, \quad \dot{\mathbf{r}}_s = \mathbf{v}_s, \quad (74)$$

with a specified screened potential, for example $V(r) = q_1 q_2 e^{-r/\lambda} / (4\pi\epsilon_0 r)$. Screening is a declared model choice, not a result of this two-body calculation. Resolve close encounters or exclude them with a stated grazing criterion. The generalized magnetized scattering theory in Ref. [14] is an important reference.

For uniform $\mathbf{B}$, compare with the unperturbed helical orbit at identical incoming conditions. Use

$$\mathbf{R}_s = \mathbf{r}_s + \frac{\mathbf{v}_s \times \mathbf{b}}{\Omega_s}, \quad \mu_s = \frac{m_s |\mathbf{v}_{\perp s}|^2}{2B}, \quad \Omega_s = q_s B / m_s. \quad (75)$$

Extract asymptotic incoming/outgoing guiding-center data at a common reference time. A raw endpoint displacement can contain an arbitrary flight-time contribution after the encounter changes parallel velocity.

Sample incident flux, impact parameters, gyrophases and relative velocities with the appropriate measure. Check timestep, start/end distance, angular sampling and screening/cutoff convergence independently. Recover a Rutherford limit before using strong-field results. Attractive close encounters outside the grazing regime are not automatically described by the reduced kinetic closure.

For independent encounters at a stated rate ν_{enc} , diagnostic drift and diffusion coefficients are

$$A_\mu = \nu_{\text{enc}} \langle \delta\mu \rangle, \quad D_{\mu\mu} = \frac{\nu_{\text{enc}}}{2} \langle (\delta\mu)^2 \rangle. \quad (76)$$

Use both, together with cross-covariances and correlation tests. A timescale estimate must specify a nonzero population scale μ_{ref} ; the quantities $\mu_{\text{ref}}/|A_\mu|$ and $\mu_{\text{ref}}^2/(2D_{\mu\mu})$ diagnose different failures of the constraint. Near $\mu = 0$, use a suitable distribution-level metric.

Compare these with an independently determined mixing time. A physically supported constrained interval requires

$$\tau_{\text{mix}} \ll t \ll \tau_{\mu}. \quad (77)$$

Choosing a small input pitch-angle rate establishes a synthetic asymptotic test, not this physical inequality. Adding two collision terms also requires a nonoverlapping process decomposition; it can otherwise double count encounters.

13 Independent homogeneous controls and encounters

Lorentz pitch-angle evolution

For a fixed physical speed shell, the implemented operator is $C_{\text{Lz}} = \nu \partial_{\xi} [(1 - \xi^2) \partial_{\xi}] / 2$, with natural endpoint flux at $\xi = \pm 1$. The Legendre equation gives eigenrates $\nu \ell(\ell + 1)/2$. Gauss–Legendre projection and exact modal exponentials provide an independent solution; the tests additionally apply the differential operator to Legendre polynomials. Thus the $\ell = 0, 1, 2, 3$ rates are $0, \nu, 3\nu, 6\nu$. Shell population and energy are preserved, while the $\ell = 1$ momentum is transferred to the fixed reservoir. The coordinate map and full Jacobian are

$$u = v\xi, \quad \mu = \frac{mv^2(1 - \xi^2)}{2B}, \quad \left| \frac{\partial(u, \mu)}{\partial(v, \xi)} \right| = \frac{mv^2}{B}, \quad d^3v = \frac{2\pi B}{m} du d\mu. \quad (78)$$

The shell comparison changes μ while preserving speed, as ordinary pitch scattering must; it is not an unbalanced bare moment diffusion term.

Nonlinear conserving Dougherty evolution

The physical three-velocity operator is

$$C_D[F] = \nu \nabla_v \cdot [(\mathbf{v} - \mathbf{U})F + \theta \nabla_v F], \quad \mathbf{U} = \langle \mathbf{v} \rangle, \quad \theta = \frac{1}{3} \langle |\mathbf{v} - \mathbf{U}|^2 \rangle. \quad (79)$$

Integration against 1, $\mathbf{v}$ and $|\mathbf{v} - \mathbf{U}|^2$ proves that $\mathbf{U}$ and θ are constant for closed homogeneous evolution. For a positive normalized Gaussian mixture with weights p_a , component means $\mathbf{U}_a$ and covariances A_a , the exact solution remains a positive mixture with

$$\mathbf{U}_a(t) = \mathbf{U} + e^{-\nu t}(\mathbf{U}_a(0) - \mathbf{U}), \quad A_a(t) = e^{-2\nu t} A_a(0) + \theta(1 - e^{-2\nu t})I. \quad (80)$$

Substituting each Gaussian into the now constant-moment Fokker–Planck equation verifies this solution; conservation makes it a solution of the original nonlinear model. It includes a non-Gaussian control, without substituting prescribed reservoir moments. Exact Gaussian gradients and Laplacians evaluate the strong operator independently of a centered temporal difference.

In `03_controls.py`, Lorentz modal errors are below 6.5×10^{-15} . The two-component Dougherty mixture has nonzero drift and anisotropic covariance. A 40^3 physical Cartesian velocity quadrature on extent 9 gives particle error 2.0×10^{-11} , covariance error 7.82×10^{-10} and centered-energy error 4.76×10^{-10} . The strong equation versus centered time derivative error is 2.21×10^{-10} . The sampled entropy increases; positive mixture reconstruction is analytic, and the smallest sampled density is 1.96×10^{-64} .

Physical three-velocity Landau weak reference

The conventional one-species normalization used here is

$$C[F] = \Gamma \nabla_v \cdot \int F F' U(\mathbf{w}) [\nabla_v \ln F - \nabla_{v'} \ln F'] d^3 v', \quad U(\mathbf{w}) = \frac{I - \hat{\mathbf{w}} \hat{\mathbf{w}}^T}{|\mathbf{w}|}, \quad \mathbf{w} = \mathbf{v} - \mathbf{v}'. \quad (81)$$

With $\int F d^3 v = n$, the SI coefficient is $\Gamma = q^4 \ln \Lambda / (8\pi \epsilon_0^2 m^2)$, matching the mass convention of Fitzpatrick's Eqs. (3.93)–(3.94) [15]. Writing $F = (n/v_*^3) \hat{F}$ gives the frequency scale $\Gamma n / v_*^3$. The Coulomb logarithm and its cutoffs are supplied parameters; this conversion does not calibrate the constrained scalar D .

For a normalized Gaussian of covariance A , $\mathbf{w}$ has covariance $2A$ and the pair score is $\mathbf{g} = -A^{-1}\mathbf{w}$. The exact weak covariance and entropy rates are

$$\dot{A} = -\frac{\Gamma}{2} \mathbb{E}[\mathbf{w} \otimes U\mathbf{g} + U\mathbf{g} \otimes \mathbf{w}], \quad (82)$$

$$\dot{S} = \frac{\Gamma}{2} \mathbb{E}[\mathbf{g} \cdot U\mathbf{g}] \geq 0. \quad (83)$$

Trace conservation follows pointwise from $U\mathbf{w} = 0$. The center velocity integrates analytically; radial/polar/gyrophase quadrature retains the full physical $d^3 w$ geometry. A separate Cartesian Gauss–Hermite expectation avoids $\mathbf{w} = 0$ by even quadrature order. A third independent check diagonalizes A and uses the Laplace integrals for $1/r$ and $1/r^3$. Define $Z(t) = \prod_j (1 + 4A_j t)^{-1/2}$ and $V_j(t) = 2A_j / (1 + 4A_j t)$; then

$$\dot{A}_i = \frac{\Gamma}{\sqrt{\pi}} \int_0^\infty Z(t) \left[t^{-1/2} \frac{V_i}{A_i} - 2t^{1/2} V_i \sum_j \frac{(1 + 2\delta_{ij})V_j}{A_j} \right] dt. \quad (84)$$

Gaussian second and fourth moments after multiplication by e^{-tr^2} give V_i and $(1 + 2\delta_{ij})V_i V_j$, proving the integrand. The substitution $t = s^2$ removes its endpoint singularity before adaptive one-dimensional integration.

For $A = \text{diag}(1.15, 1.15, 0.7)$ and $\Gamma = 1$, the covariance derivatives are

$$(-0.0740393, -0.0740393, 0.148079), \quad \dot{S} = 0.0413884.$$

The reported energy rate residual is 2.1×10^{-18} . Radial quadrature refinement reaches a 4.14×10^{-14} discrepancy from (84) at order 28, while the independent Cartesian Hermite route converges more slowly from 8.83% at order 8 to 0.316% at order 48. Tail and denominator-softening scans are saved separately. Softening changes only $1/r$ and retains $U\mathbf{w} = 0$; it is a convergence diagnostic, not an unmodified Coulomb tensor. A small-anisotropy check takes $A = I + \epsilon \text{diag}(1, 1, -2)$. At first order, $\mathbb{E}|\mathbf{w}| = 4/\sqrt{\pi}$ for covariance $2I$ and the angular moments are $\mathbb{E}[n_i^2] = 1/3$ and $\mathbb{E}[n_i^2 n_j^2] = (1 + 2\delta_{ij})/15$. Substitution in the weak covariance rate gives $\dot{A}_i = -4\Gamma \epsilon D_i / (5\sqrt{\pi}) + O(\epsilon^2)$, where $D = (1, 1, -2)$. The instantaneous normalized stress rate approaches $4\Gamma / (5\sqrt{\pi}) = 0.451351667$; for $\epsilon = 0.03, 0.01, 0.003$ the two independent references agree and give 0.457605, 0.453335, 0.451936 respectively. This is a limiting conventional control, without a discovery claim.

Maxwellian score differences are parallel to $\mathbf{w}$ and annihilate the kernel for multiple drifts and temperatures. A separate positive non-Gaussian nodal pair calculation checks weak number/momentum/energy conservation and positive entropy production. Its omitted coincident-node contribution is a stated quadrature prescription, not a general directional limit. These controls verify instantaneous weak moments; no full nonlinear Landau kinetic time integration or Gaussian closure trajectory is claimed.

Direct screened encounter audit

`04_encounters.py` directly integrates equal-mass, equal-charge repulsive pairs in a uniform field with $V(r) = \kappa e^{-r/\lambda}/r$. It measures changes relative to the free helix at identical incoming conditions. Endpoint refinement holds fixed the incoming guiding center and phase at the $z = 0$ crossing; changing the initial plane at fixed raw position and phase would instead change the orbit and is not a valid endpoint convergence test. For relative guiding-center offset b , $v_{\parallel}$ and gyrophase φ at that crossing, the incoming plane $z = -L$ has phase $\varphi + \Omega L/v_{\parallel}$ and transverse position $(b - v_y/\Omega, v_x/\Omega)$. This follows directly from the invariant free $\mathbf{R} = \mathbf{r} + \mathbf{v} \times \mathbf{b}/\Omega$.

A documented case uses $b = 1.2$, $v_{\parallel} = 1$, $v_{\perp} = 0.35$, $\Omega = 2$, $\kappa = 0.05$, $\lambda = 4$ and 32 phases. Increasing L from 12 to 32 changes the mean moment increment from 5.0126×10^{-5} to 5.0074×10^{-5} and its second moment from 9.4135×10^{-7} to 9.4059×10^{-7} . The individual incoming μ is 0.00765625, so the rms increment is approximately 12.7% of that scale while the ensemble pair-energy error is below 3.4×10^{-11} . This is a controlled counterexample to inferring moment preservation solely from energy conservation; it does not characterize every weak encounter regime. For the same $B = 2$ case, the maximum relative velocity change from the free helix is 0.0340, the minimum sampled separation is 1.045, and $\rho/\lambda = 0.0219$, $\rho/b_{90} \simeq 0.982$, $\Omega\lambda/v_{\text{rel}} \simeq 7.55$. Uniform B gives $\rho/L_B = 0$; ν/Ω is unassigned because no encounter rate is measured. The free-orbit control has moment drift below 1.2×10^{-14} and guiding center drift below 6×10^{-16} . In a weak-field unscreened Rutherford control, increasing endpoint distance from 12 to 96 reduces the angular error, but the final relative discrepancy remains 1.11×10^{-3} and is recorded as finite-endpoint error. A separate close-encounter timestep scan decreases pair-energy error from 5.40×10^{-4} to 3.57×10^{-9} . The two individual moment increments are correlated by the imposed symmetric center-of-mass initialization; saved joint covariances must not be interpreted as independent species statistics.

No incident-flux/impact-parameter ensemble, plasma encounter frequency, or independently measured lifetime is assigned to these phase statistics.

14 Research questions and evidence

Geometry and nullspace. Begin with the proven finite-quadrature spectrum (33). Track its changes under a valid mirror, toroidal or dipole field. Separate collision null modes from decay generated through coupling to Hamiltonian transport. A small positive eigenvalue on one finite grid is not a continuum gap. Vary all resolutions, domains and velocity tails.

Localization. Compare the local kernel with a specified symmetric finite-range kernel, using a derived normalization as the range shrinks. Boundary renormalization must retain pair symmetry. A Gaussian-range surrogate remains a surrogate unless independent scattering physics supports it.

Constrained dipole states. Ref. [16] studies maximum-entropy dipole states and additional adiabatic constraints. Here the result sought is a dynamical relaxation law, an extra invariant, a loss of mixing, or an independently measured lifetime. Reproducing (38) is verification rather than discovery.

For any geometry-amplitude scaling, compare competing fits and inspect projector degeneracies before using regular perturbation theory. For steady-state parameter derivatives specify which particle, energy and marginal constraints are held fixed; otherwise a nullspace can make the derivative underdetermined.

Executed prototype evidence and remaining limits

The public library has five substantive modules: `geometry`, `collisions`, `solver`, `controls` and the independent `reference`. Examples save inputs, boundary/model descriptions, package versions, source checksums and commit context with their results. All quoted values below are normalized, finite-grid evidence from the saved runs; rates are not dimensionally calibrated to a plasma.

The earlier 216-node NumPy oracle verifies the local uniform rate 0.11742603550295856 and its exact velocity-nullspace structure. The JAX implementation is compared with that independent pair calculation, a dense weak assembly, analytic Fourier decay, nonlinear directional linearization, conservation and discrete entropy identities. Geometry tests additionally cover both charges, both coordinate charts, sampled Jacobi and Liouville identities, independent finite-difference convergence, measure transformation, electric cross-field drift and mirror orbit timestep convergence.

`01_relaxation.py` evolves a positive 105-node uniform nonlinear state for ten steps of $\Delta t = 0.08$. Its saved particle, energy and maximum moment-bin errors are below 7.1×10^{-17} ; the minimum step entropy increase is 4.75×10^{-4} and the maximum discrete entropy identity error is 6.08×10^{-15} . Nodal values remain positive, and the continuous spatial Fourier certificate and positive piecewise-linear velocity reconstruction are checked by the example. This is the specified nonlinear local surrogate, without self-consistent fields or a claim that its additional null directions disappear.

`02_geometry.py` evolves collision-only, ideal-only and combined linear states on the nonuniform vacuum toroidal annulus. On the base $5 \times 3 \times 3 \times 11 \times 13$ grid at $t = 0.24$, the respective $Q/Q(0)$ values are 0.820926, 1.000000 and 0.810980. Number, energy and moment-bin errors are below 1.1×10^{-17} for these three controls. Three-level scans separate radial, angular, parallel-velocity, moment and timestep errors; velocity tails are varied separately. The last changes in both final Q and dissipated fraction pass the 1% target for every scan. Changes in the dissipated fraction are 0.332% radially, rounding-level in the two angular grids for this resolved initial mode, 0.709% in parallel velocity, 0.638% in moment resolution, 0.495% in time, 0.0578% in parallel-velocity tail extent, and 0.519% in moment-tail extent. These are separate refinements of a specified observable, not a mathematical continuum convergence theorem. The unmodified 243-node collision matrix has 39 near-null eigenvalues, compared with the 39-dimensional independent invariant span from spatial populations and angular-weighted invariants in (53), with scaled eigenpair residual 1.63×10^{-15} . Every observed tiny-grid null direction is accounted for by that span; the eigenvalues themselves are unchanged. No continuum gap or uniqueness follows from that finite matrix.

`06_range.py` checks 27 combinations of spatial grid, velocity count and Gaussian range. The finest spatial density-rate discrepancy is at most 3.03×10^{-13} relative to (35). Underresolved narrow ranges are marked unresolved: for $\ell = 0.15$, 16 spatial nodes give a 58.5% density-rate error, while 32 give 4.75×10^{-4} and 64 reach rounding accuracy. This explicitly demonstrates why the range must be resolved before interpreting density-nullspace lifting.

`07_equilibria.py` independently integrates the stationary density (38) in all five fields with three velocity quadrature orders and two tail extents. The largest finest error on $|u| \leq 7$, $\mu \leq 20$ is 1.61×10^{-7} (dipole). Its separate full-marginal finite-quadrature energy solve gives $\beta = 1.7310695$, zero reported energy residual and moment-bin residual 2.22×10^{-16} . This verifies candidate construction, not attraction from arbitrary initial data.

`08_fields.py` performs actual nonlinear collision-only steps in mirror, dipole and perturbed-dipole boxes with the bounds derived above. At three spatial resolutions, equilibrium residuals are below 2.8×10^{-14} and initial entropy production is positive. The last spatial changes in initial production are 0.336%, 0.255% and 0.257% respectively. Three positive nonlinear steps on each coarse box conserve number, energy and the full moment marginal to below 2.0×10^{-17} in the saved run, with positive

entropy increments. Velocity resolution/tails and longer nonlinear time evolution remain unresolved for these boxes. These runs impose natural collision flux only; they do not impose a confined dipole Hamiltonian boundary or demonstrate dipole self-organization.

Independent controls and direct encounters are described in the preceding sections. Performance measurements separate first-call compilation, warmed synchronization, pair chunk size and solver residual. Implicit parameter derivatives are checked against dense solves and centered finite differences. All performance claims remain specific to the saved device and accuracy; no unmeasured GPU speedup is claimed. The selected P1–P5 prototype examples have executable structural and convergence evidence. This does not certify every geometry or parameter regime; joint box time-evolution convergence and distinct-state zero-set extensions remain open. Physical coefficient calibration, unequal-species closure, self-consistent electrostatics, and measured plasma mixing-versus-moment timescale separation are not implemented claims.

Directional zero-set check

At the off-axis dipole state $(1, 0.2, 0.3, 0.6, 0.4)$, the independent `test_orbits.py` calculation approaches coincidence along u and μ separately at separations $10^{-2}, 10^{-3}, 10^{-4}$. The two limiting $PI_X P$ tensors remain different: there is no direction-independent continuous projector extension. For the smooth observable $x^2 + u^3 + \mu x$, however, $\Delta\phi^T PI_X P \Delta\phi$ decreases quadratically along both paths. Thus identical-node diagonal contributions vanish in the weak form despite the kernel’s directional discontinuity. This fact does not authorize deletion of distinct states with the same energy-flow vector; the general constructor rejects those ambiguous pairs. The uniform directional prescription is derived separately and tested against the exact oracle.

Measured costs at matched error

For four independent uniform Fourier blocks with 128 velocity nodes each, $\Delta t = 0.025$ and $\lambda = 0.3$, the common temporal error is 2.77×10^{-5} . On the Apple M4 CPU with JAX 0.11.2 and SOLVAX 0.28.2, five synchronized repetitions give the following measured values:

Route	Compile/first (s)	Median (μs)	Range (μs)	Temporary bytes
Dense	0.092	550.3	214.2–1297.8	262852
PCG	0.184	571.7	273.1–804.7	16696
Diagonal PCG	0.225	43.8	34.5–126.6	16832

The implicit solve residuals are below 6×10^{-13} ; these separable uniform timings do not describe general pair-kernel complexity. For 945 actual weak pairs, a 16-pair chunk reduces XLA temporary buffers from 118913 to 16480 bytes, while the warm median increases from 96.3 to 158.2 μs . The complete benchmark process peaks near 332 MB RSS, including the runtime and compiled executables. The base combined toroidal evolution takes 2.38 s for six steps after 1.12 s compilation/first execution and 1.22 s setup. Its state is 51480 bytes and its stored pair list contains 456885 unordered pairs; pair-application chunk workspace and all version/input/commit details are in the saved JSON. No extrapolated speedup or isolated-RSS interpretation is asserted.

Nonuniform velocity quadrature and resolved tails

The additional `11_field_velocity.py` campaign evaluates the initial nonlinear entropy production on the same mirror, dipole and nonaxisymmetric boxes, with fixed 3^3 spatial quadrature. Its positive initial distribution is

$$f = \exp[-E - 0.2\mu + 0.1 \sin(x)u + 0.04y\mu]. \quad (85)$$

Positive Gauss–Legendre velocity quadrature replaces both the population weight $w_X Bw_u w_\mu$ and the local pair weight $w_X (Bw_u w_\mu)(Bw'_u w'_\mu)$ consistently. Derivatives and energy directions are unchanged. The three-point derivative is exact for the quadratic velocity energy and the linear- u logarithmic perturbation on either node family. Number and energy are checked independently using shifted Gaussian integrals in u and exponential integrals in μ .

The base velocity domain is $u \in [-4, 4]$, $\mu \in [0, 20]$, with orders 25 and 21. Separate order scans are 17, 21, 25 in u and 13, 17, 21 in μ . Tail extensions to $|u| = 5$ and $\mu = 24$ are separately checked at wider-domain orders 29 and 25. A Gauss tail extension changes node spacing; these extra order checks are essential and are not described as fixed-spacing tests. All 36 cases retain their inputs and status. Across all three fields, the largest final Gauss production change is 0.261%; the largest expanded-domain quadrature change is 0.180%. Base finite-domain number/energy quadrature errors are below 9.4×10^{-15} . Independent analytic omitted energy is about 0.050%, which is not itself a bound on production error.

Equidistant trapezoid controls at $9^2, 17^2, 25^2$ velocity nodes remain unresolved: the last production changes are 16.63%, 10.14%, and 10.15% for mirror, dipole and nonaxisymmetric fields. Their statuses are preserved. The successful Gauss checks establish the stated initial-production comparison on a fixed spatial grid, not joint spatial–velocity or finite-time nonlinear convergence. The constructor now rejects zero pair directions without allocating and discarding a 25-component tensor per pair; the retained unit directions and collision action are unchanged and tested against explicit projector matrices.

Bounded Maxwellian encounter flux and thermal center motion

Let $\theta = k_B T/m$ denote a single-particle velocity variance. For two equal Maxwellian particles, $\mathbf{w} = \mathbf{v}_1 - \mathbf{v}_2$ and $\mathbf{V} = (\mathbf{v}_1 + \mathbf{v}_2)/2$ are independent, with covariances $2\theta I$ and $\theta I/2$. The transformation has absolute Jacobian one. The relative distribution and equations for a screened pair potential are

$$F_{\text{rel}}(\mathbf{w}) = (4\pi\theta)^{-3/2} e^{-|\mathbf{w}|^2/(4\theta)}, \quad (86)$$

$$\dot{\mathbf{r}} = \mathbf{w}, \quad \dot{\mathbf{w}} = \Omega \mathbf{w} \times \mathbf{e}_z + \frac{2\kappa e^{-r/L_s} (1 + r/L_s)}{m r^3} \mathbf{r}, \quad \Omega = qB/m. \quad (87)$$

The conserved relative energy is $m|\mathbf{w}|^2/4 + \kappa e^{-r/L_s}/r$; center motion follows an independent free helix. The new six-dimensional relative integrator is checked against the independent twelve-dimensional pair integrator, including nonunit masses and both charge signs.

For guiding-center impact b , the translation to the incoming perpendicular position has unit Jacobian. Axisymmetry integrates its orientation to $2\pi b db$. On a plane normal to the field, the two-sided incoming crossing measure per partner density is

$$\frac{d\nu}{n} = 2(2\pi b db) w_{\parallel} F_{\text{rel}}(\mathbf{w}) w_{\perp} dw_{\perp} dw_{\parallel} d\varphi, \quad w_{\parallel} > 0. \quad (88)$$

The factor two includes the opposite incoming direction. There is no unordered-pair half factor for this tagged-particle flux, and the plane flux uses $|w_{\parallel}|$, not the cross-section flux $|\mathbf{w}|$. For impact area $A = \pi(b_{\text{max}}^2 - b_{\text{min}}^2)$, the bounded integral is $2A(2\pi)J_{\parallel}J_{\perp}/(4\pi\theta)^{3/2}$, where $J(a, b) = 2\theta[e^{-a^2/(4\theta)} - e^{-b^2/(4\theta)}]$. This supplies an independent analytic normalization.

Undo outgoing free gyromotion when defining $\Delta \mathbf{w}_\perp = \overline{\mathbf{w}}_{\text{out},\perp} - \mathbf{w}_{\text{in},\perp}$. With $c = m\Delta|\mathbf{w}_\perp|^2/(8B)$, the individual changes are $\Delta\mu_{1,2} = c \pm [m/(2B)]\mathbf{V}_\perp \cdot \Delta\mathbf{w}_\perp$. Exact Gaussian center averaging therefore gives

$$\langle \Delta\mu_1 \rangle_V = c, \quad (89)$$

$$\langle (\Delta\mu_1)^2 \rangle_V = c^2 + \sigma_c^2, \quad \langle \Delta\mu_1 \Delta\mu_2 \rangle_V = c^2 - \sigma_c^2, \quad (90)$$

$$\sigma_c^2 = \frac{m^2\theta}{8B^2}|\Delta\mathbf{w}_\perp|^2. \quad (91)$$

A separate Gaussian quadrature of the actual individual initial/final velocities verifies these raw moments. A zero-center trajectory omits σ_c^2 . The drift and diffusion moment integrals are $A_\mu/n = \int \langle \Delta\mu \rangle_V d\nu/n$ and $D_{\mu\mu}/n = \frac{1}{2} \int \langle (\Delta\mu)^2 \rangle_V d\nu/n$; the latter half is the second-moment convention, not pair counting.

The executed normalized case uses $m = q = 1$, $B = 2$, $\kappa = 0.03$, $L_s = 3$, $\theta = 0.5$, $b \in [1.2, 2.4]$, $w_\parallel \in [0.7, 1.5]$ and $w_\perp \in [0.2, 0.8]$. The $4^3 \times 16$ base quadrature is independently refined in each impact/speed coordinate through orders 3, 4, 5, in gyrophase through 8, 16, 32, and by a half-grid phase shift. Moving the planes through $L = 18, 24, 32$ preserves the same incoming free helix. The first outward exit is recorded; reflected trajectories are retained, and failure to exit within $6L/w_\parallel$ raises. Eight held-out states additionally check active maximum steps 0.2, 0.1, 0.05. Direct quadrature changes are below 0.021%; analytic flux normalization error is 1.62×10^{-9} . The bounded results are $\nu/n = 1.2601061$, $A_\mu/n = 3.11546 \times 10^{-6}$ and $D_{\mu\mu}/n = 8.00765 \times 10^{-7}$ in the stated normalized units.

A supplementary multilinear scattering table is tested on 64 uniformly sampled states in the inner quadrature-node hull (seed 701). Its held-out second-moment relative RMS error is 20.5%, exceeding the predeclared 10% target; this surrogate remains unresolved despite converged direct integrals. These speed bands contain only 11.6% of the full thermal incoming flux in the same annulus. Omitted impact/speed regions and successive encounter correlations remain untested. Truncation can exclude reverse encounters, so the positive conditional drift is not a full Maxwellian-plasma drift. Neither these bounded crossing coefficients nor the interpolation checks calibrate the constrained scalar D or establish a dipole lifetime.

Reproducible evolution visualization

The README animation is a solution of the uniform linear surrogate at finite velocity quadrature, not a nonlinear encounter or physical-rate simulation. The nine positive velocity masses M_j are proportional to $\exp(-v_j^2/2)$, with v_j equally spaced on $[-2, 2]$ and $\sum_j M_j = 1$. Define

$$p_j = \frac{v_j^2 - \sum_i M_i v_i^2}{\max_i |v_i^2 - \sum_l M_l v_l^2|}, \quad h_j(x, 0) = (0.25 + 0.20p_j) \cos x. \quad (92)$$

The density and velocity-neutral amplitudes are evolved using matrix exponentials of an independently assembled ordered-pair Fourier matrix ($N_x = 64$, Gaussian range $\ell = 0.6$) and the analytic local matrix. For $D = 0.8$, $B = 1.7$, $q = -1.3$, and $\tau = \lambda t$, the exact comparisons are

$$h_{j,\text{local}}(x, t) = [0.25 + 0.20e^{-\tau} p_j] \cos x, \quad (93)$$

$$h_{j,\text{range}}(x, t) = [0.25e^{-(1-e^{-0.6^2/2})\tau} + 0.20e^{-\tau} p_j] \cos x. \quad (94)$$

All 65 saved frames on $0 \leq \tau \leq 8$ are checked against these formulas; $1 + h_j$ stays positive. Fixed color limits prevent apparent relaxation caused by automatic rescaling. The displayed velocity coordinate is a discrete node index, not a physical three-dimensional velocity grid. GIF and MP4 use the same frames, and the arrays are saved in a compressed archive. The accompanying validation and cost panels read the existing experiment JSON files. Their metadata retain each source file's hash and source commit, rather than attributing old measurements to the visualization revision.

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